



ORCA Documentation

High Fidelity Physical AI Simulation System

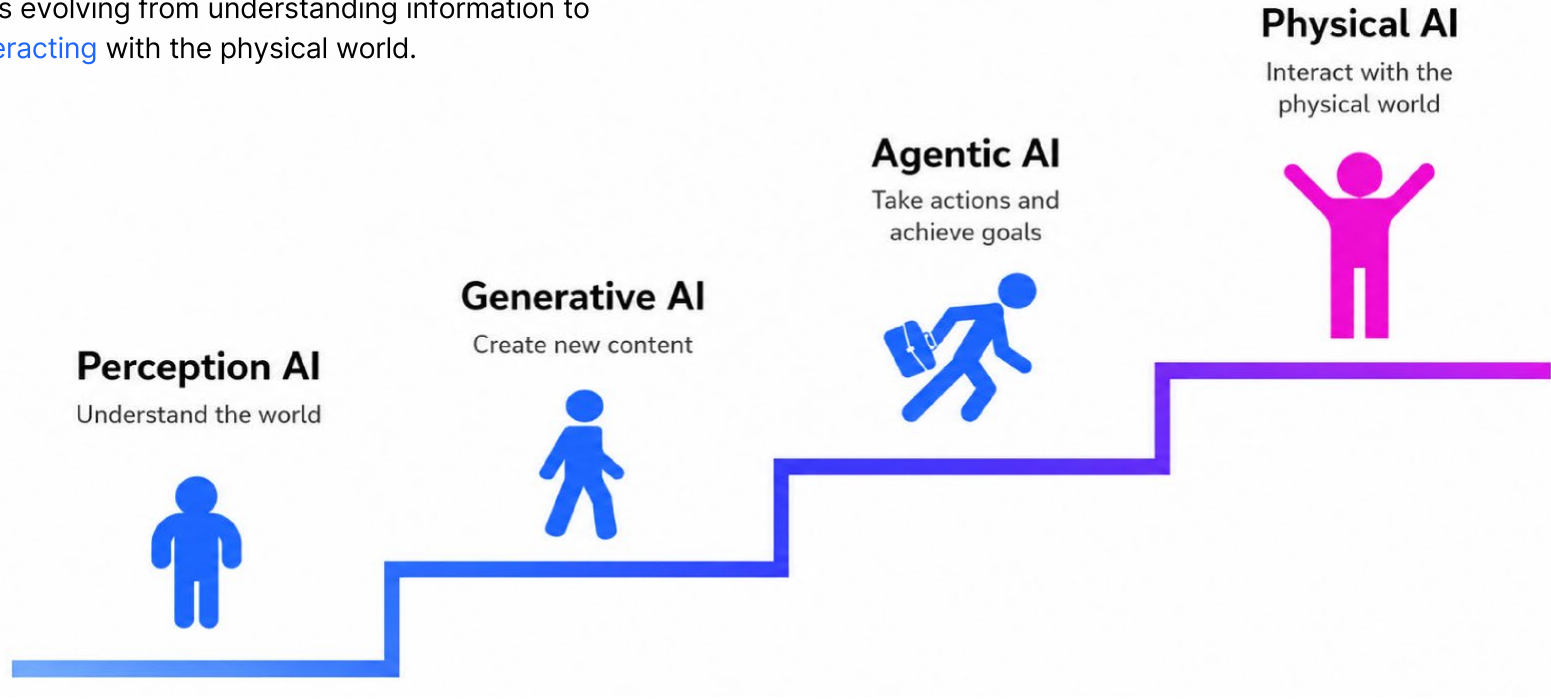


Part 1

Let's start by talking about Physical AI

Evolution of Artificial Intelligence

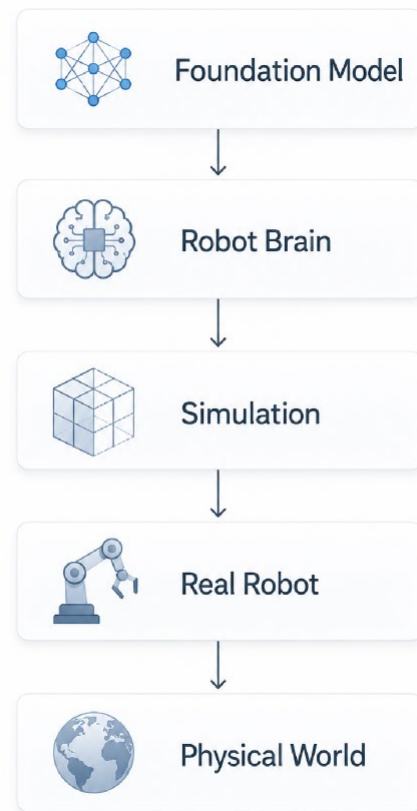
AI is evolving from understanding information to [interacting](#) with the physical world.



What is Physical AI?

Physical AI enables AI models to **perceive, understand and interact** with the physical world through simulation and real-world deployment.

Unlike traditional AI that processes text or images, Physical AI focuses on embodied intelligence and robot behaviors.



Why Physical AI?

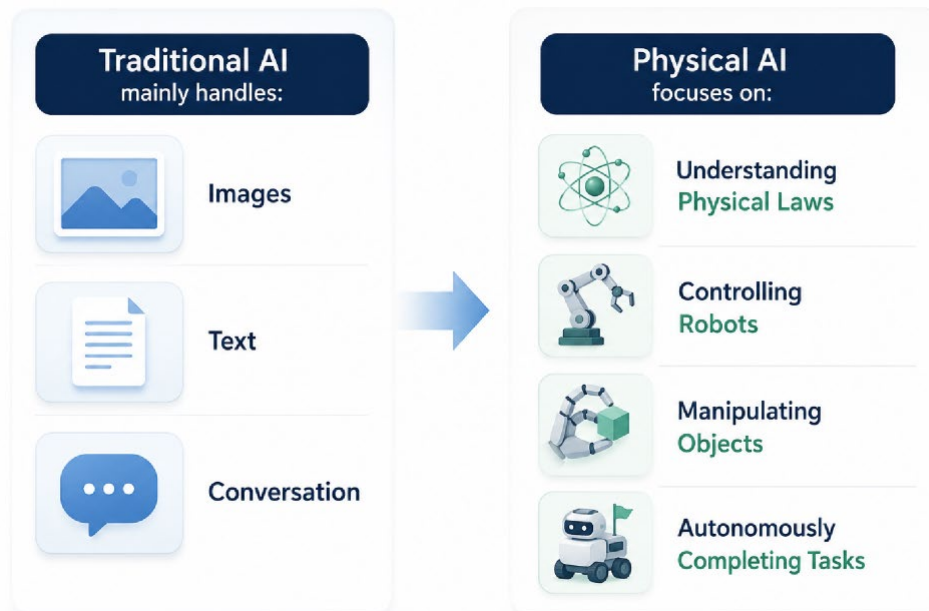
Traditional AI

- Understands images
- Generates text
- Answers questions

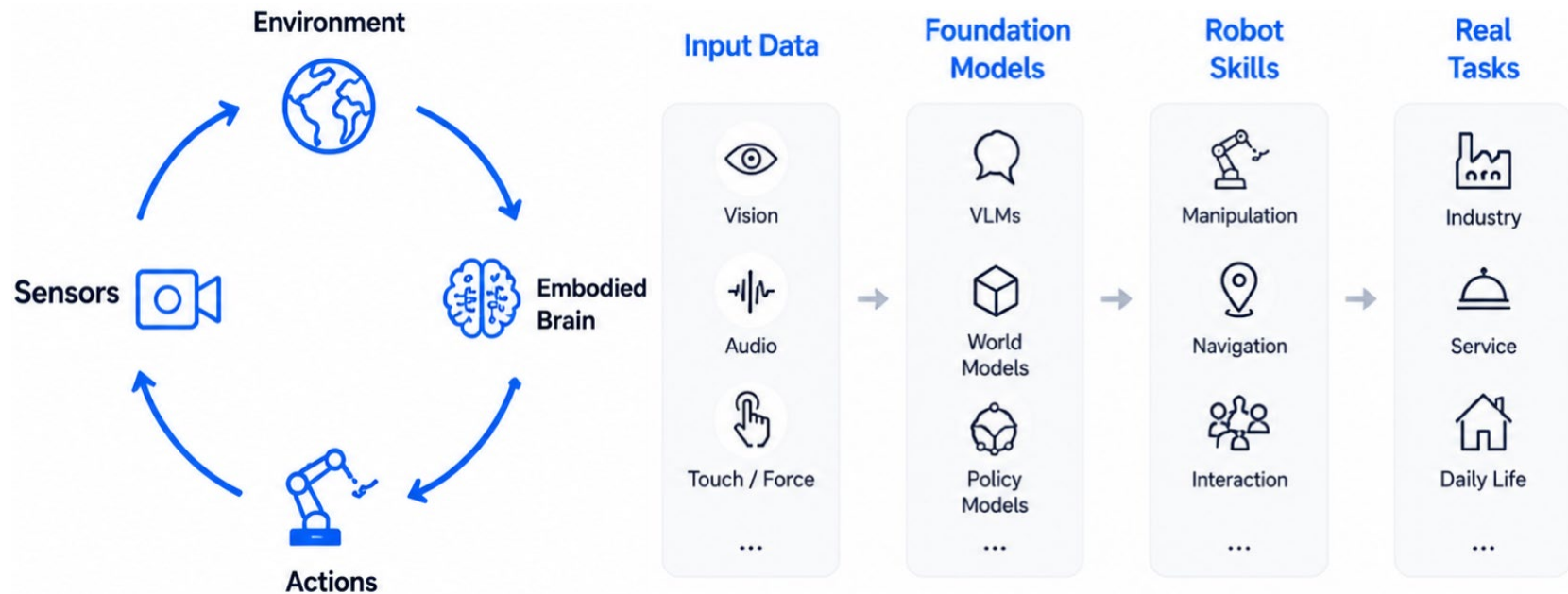
VS.

Physical AI

- Understands physics
- Controls robots
- Learns manipulation
- Performs autonomous tasks



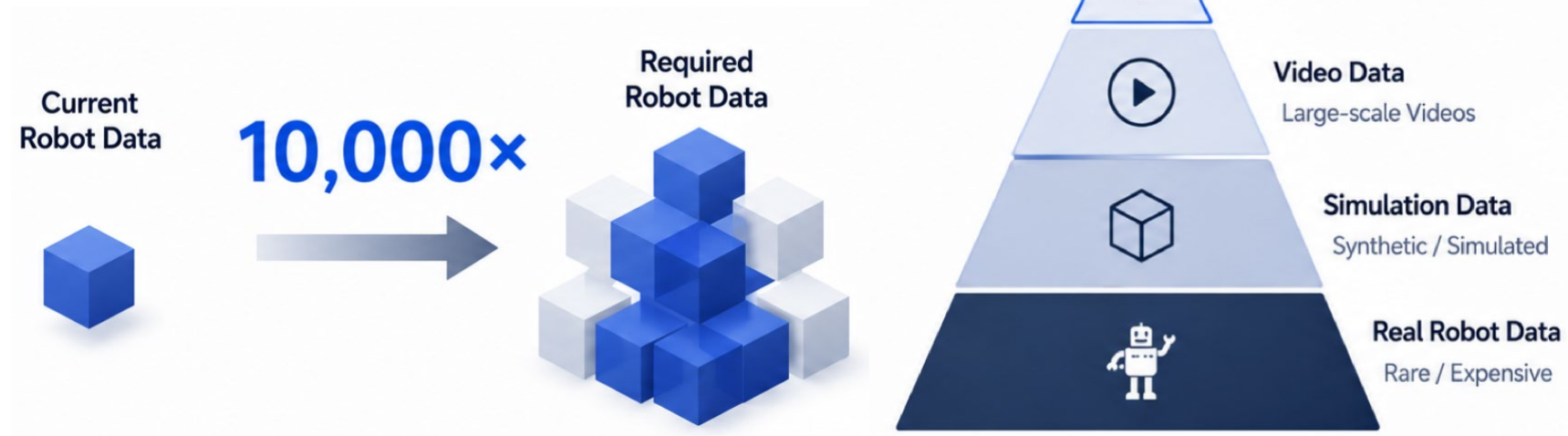
Embodied Intelligence is Powered by Data



Data is the fuel that drives Embodied intelligence.

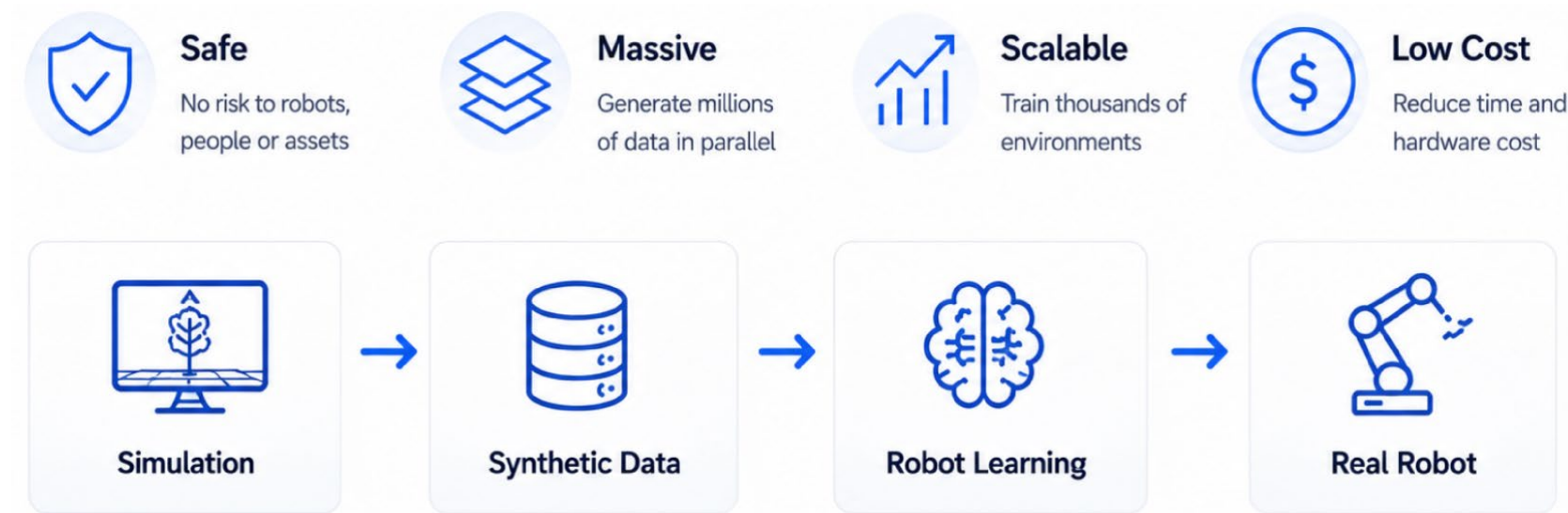
The Data Bottleneck of Physical AI

Why is Physical AI **harder** than Generative AI?



Real-world Embodied Data is scarce, expensive, and difficult to collect.

Why Simulation?



Simulation bridges the gap between data scarcity and Embodied intelligence.

Real World



High Cost

Expensive equipment, maintenance and data collection



Safety Risks

Potential damage to robots, people and environment



Slow Iteration

Limited trial-and-error, long development cycle



Limited Data

Hard to collect diverse and large-scale data



Low Reproducibility

Results vary due to uncontrollable factors

Vs

Simulation



Massive Data

Generate unlimited, diverse training data



Parallel Training

Run thousands of environments in parallel



Safe

No risk, free to explore and make mistakes



Repeatable

Fully controllable and highly reproducible



Customizable

Easily adjust scenarios, sensors and parameters

Train in simulation, deploy in reality.

Part 2

Simulation is the only scalable way to bridge this gap
Now is the time to try ORCA Physical AI Simulation System

ORCA Lab

It enables users to build simulation scenes, import robots, generate synthetic data, and validate AI models in one unified platform.

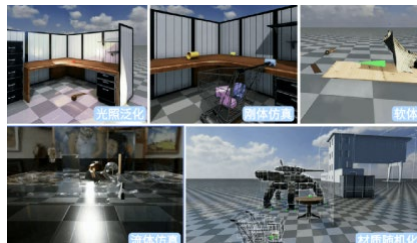
“Say goodbye to complex code, robot training for everyone”





Songying OrcaLab Simulation System

Building a physically accurate, multi-scenario, multi-form embodied robot simulation platform



High precision physical rendering & simulation

- Create a physical AI training environment
- High-fidelity PBR rendering
- Real-time physics simulation like rigid bodies and fluids

100+

Full-scenario, multi-modal high-precision simulation training environment

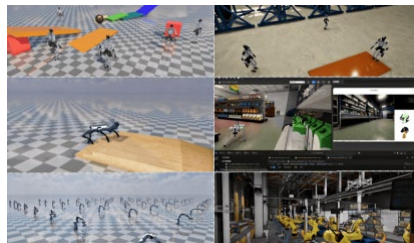


Efficient parallel training

- Multi-robot parallel control/training
- Supports multiple control strategies like MPC and WBC
- RL reinforcement learning, 4096 instances high concurrency

100 times

Speeding up embodied AI large model training



Full perception for robots

- Multiple sensors (RGB, Depth, IMU, Lidar)
- Multi-view adjustment control
- Multi-dimensional force sensing, tactile sensing

100+

Achieving millimeter precision interactive simulation based on physical laws



Synthetic Simulation Data & Training & Evaluation

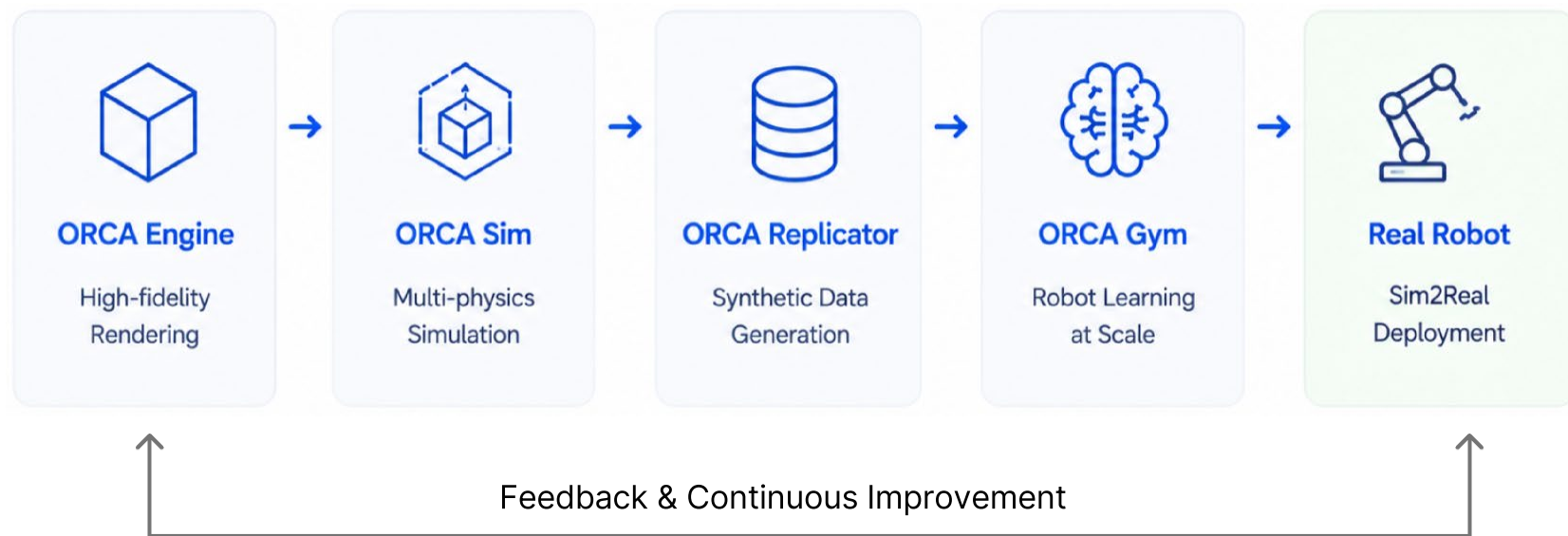
- Provide vertical, high-precision synthetic data
- Support full-link data collection, data playback, data augmentation, and data visualization.

100 times

Accelerating data collection for robot interaction synthesis

ORCA Physical AI Simulation System

ORCA provides a complete [Physical AI infrastructure](#) for [simulation](#), [synthetic data generation](#), [robot learning](#), and [Sim2Real deployment](#).



ORCA Gym: Reinforcement Learning

The robot performs actions in a simulated environment and keeps adjusting its behavior based on the **observations** and **rewards** it gets from the **environment**, aiming to maximize its total rewards.

Standard RL Environment

- Gym-compatible reinforcement learning interface
- Flexible task and reward design
- Easy integration with custom environments

Massive Parallel Training

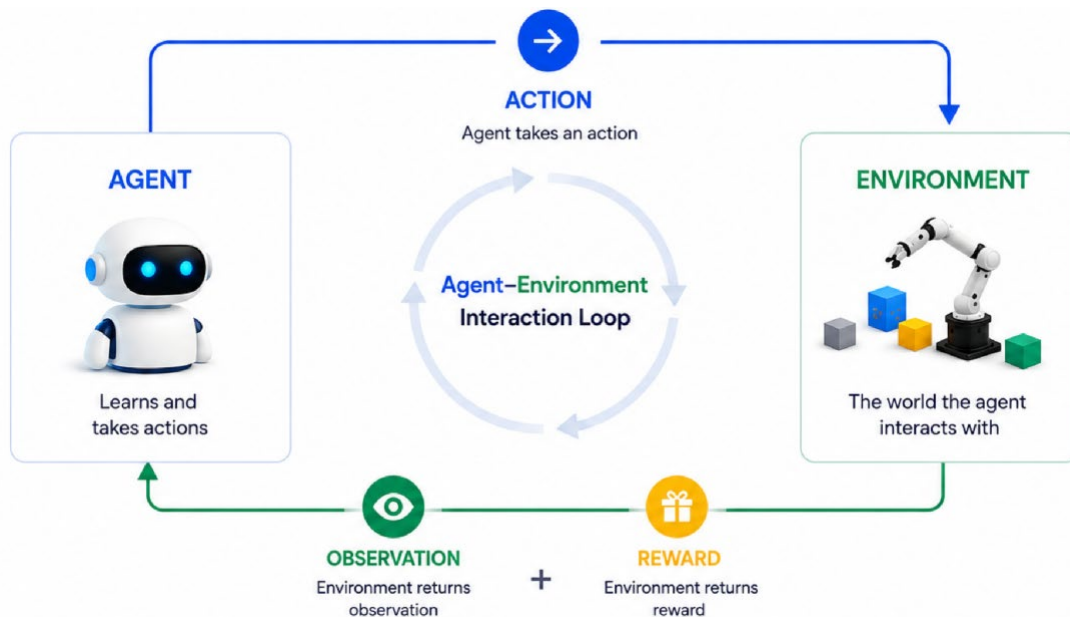
- GPU-accelerated simulation
- Thousands of environments running simultaneously
- Significantly improves training efficiency

Flexible Algorithm Integration

- Compatible with mainstream RL frameworks
- Supports custom training pipelines
- Easy policy evaluation and iteration

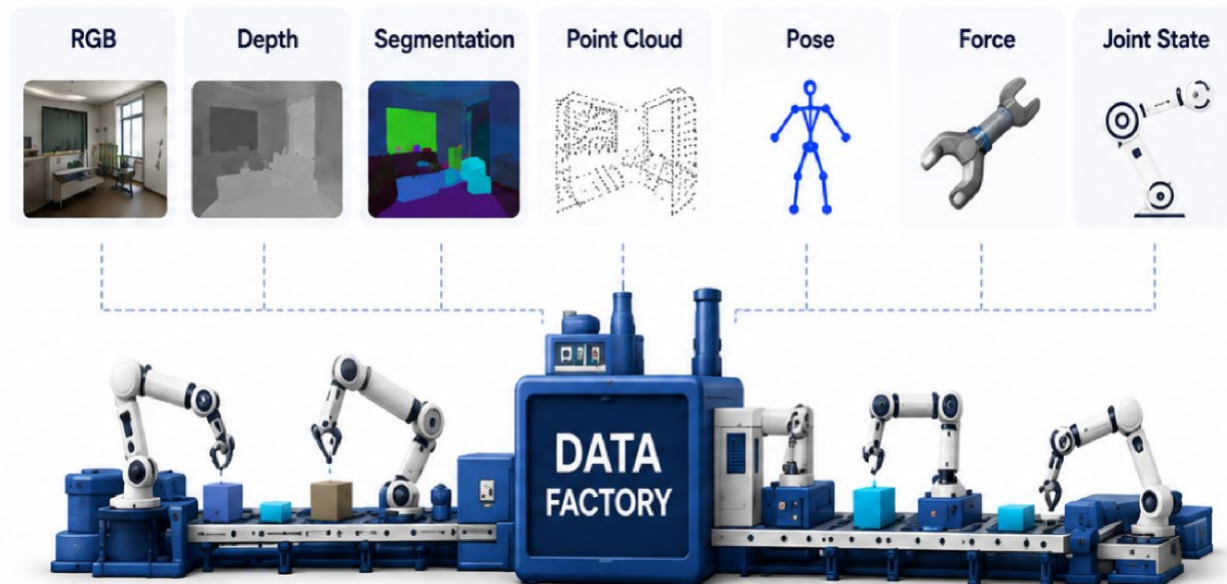
Sim2Real Ready

- Physics-based validation
- Sensor simulation
- Domain randomization for robust policy transfer



Synthetic Data: Generate Unlimited Data

Multi-model Data Output



Generated Automatically. No Manual Annotation.



Domain Randomization

Randomize textures, lighting, objects, and physics



Sensor Simulation

Simulate cameras, LiDAR, IMU and more



Automatic Annotation

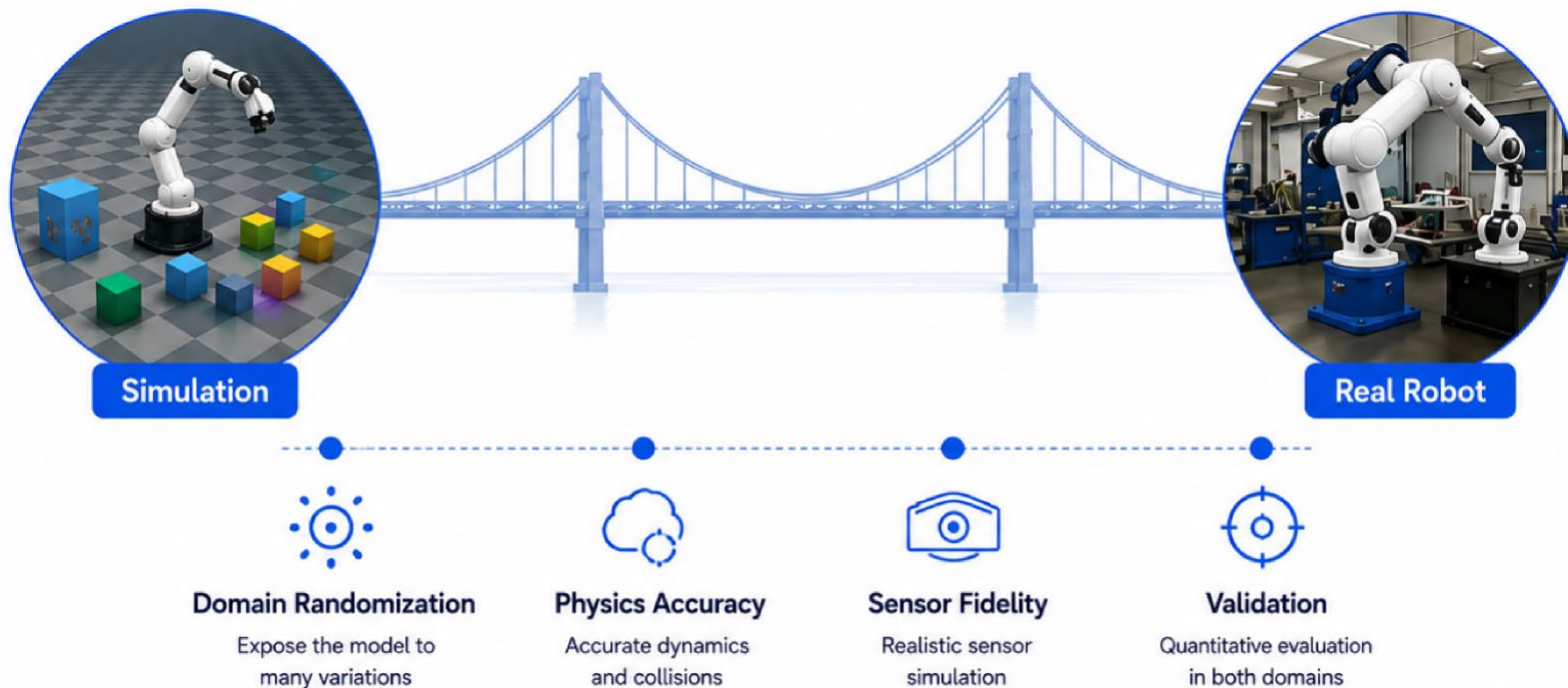
Labels, 3D bounding boxes, poses and more



Scalable Generation

Generate billions of samples efficiently and continuously

Sim2Real: Bridging Simulation and Reality



The closer the simulation is to reality, the better the transfer performance.

Step 1

ORCA Lab Installation

1. First, you need to download the OrcaLab EXE installer. The download link is below:
<https://www.orca3d.cn/download.html>
2. Then, please double-click the downloaded EXE installation file and follow the installation wizard to complete the installation.
3. Next, you can launch OrcaLab via the desktop shortcut or the Start Menu.

Note that the first launch will automatically install the necessary dependency packages. Just wait for the installation to complete, which may take some time.

Latest Version

Download ORCA Lab

Download ORCA Lab for Windows

For Linux version, please refer to the product documentation for installation.

Install ORCA Lab for Linux

ORCA Lab Version Information

ORCA Lab is an open-source physics AI simulation platform designed for robot training, autonomous driving simulation, and digital twins.

Version	Python Version	Download Link	Initial Release	End of Support
v26.6.3 Latest	Python 3.12	Download Windows 64-bit	June 30, 2026	
v26.5.1	Python 3.12	Download Windows 64-bit	May 29, 2026	August 27, 2026

Step 1

ORCA Lab Installation

If you run into some issues while installing the EXE, or you want to deploy it not just on Windows but also on Ubuntu, please follow the link below for more detailed installation instructions. It provides very detailed guidance on how to do pip installation, set up the required environment, and deploy on Windows or Ubuntu systems. At the same time, you can also explore more content here.

Please note, if you choose to install via pip, OrcaLab will start using the command line. Each time, you need to activate the conda environment before starting OrcaLab.

<https://docs.orca3d.cn/en/#/README>

The screenshot shows the ORCA Lab Documentation website. The left sidebar contains a navigation menu with the following items: ORCA Lab, Welcome to ORCA Lab, Product Positioning, Core Advantages, Target Audience, Product Format, Environment Setup, User Registration & Management, Linux Installation Guide (highlighted with a red box), Windows Installation Guide (highlighted with a blue box), Quick Start Simulation Guide, User Guide, Basic Operation Guide, Asset Library Operation Guide, Data Collection Guide, Developer Guide, OrcaGym API Documentation, MCP & CLI Operation Guide, FAQ & Troubleshooting, and FAQ. Two red arrows point from the 'Linux Installation Guide' and 'Windows Installation Guide' links to a red box labeled 'Ubuntu installation guide'. A blue arrow points from the 'Windows Installation Guide' link to a blue box labeled 'Windows installation guide (include pip/exe installation and how to set up the required environment)'.

Search

ORCA Lab

ORCA Lab Documentation

ORCA Lab

Welcome to ORCA Lab

- Product Positioning
- Core Advantages
- Target Audience
- Product Format

Environment Setup

User Registration & Management

Linux Installation Guide

Windows Installation Guide

Quick Start Simulation Guide

User Guide

- Basic Operation Guide
- Asset Library Operation Guide
- Data Collection Guide

Developer Guide

- OrcaGym API Documentation
- MCP & CLI Operation Guide

FAQ & Troubleshooting

FAQ

Welcome to ORCA Lab!

Welcome to the official documentation for the ORCA Lab, including guides, and other core content.

Product Positioning

ORCA Lab is a high-fidelity physics AI simulation system designed for research faculty and students, individual users, and non-commercial organizations. It offers a Free Edition, allowing users to quickly experience the full core capabilities.

Core Advantages

- **Ease of Use:** No complex coding background required. Supports one-click installation and training.
- **Full Workflow Coverage:** From physics AI simulation environment setup to simulation training.
- **Flexible Adaptation:** Supports personalized operations and deployment.

Product Format

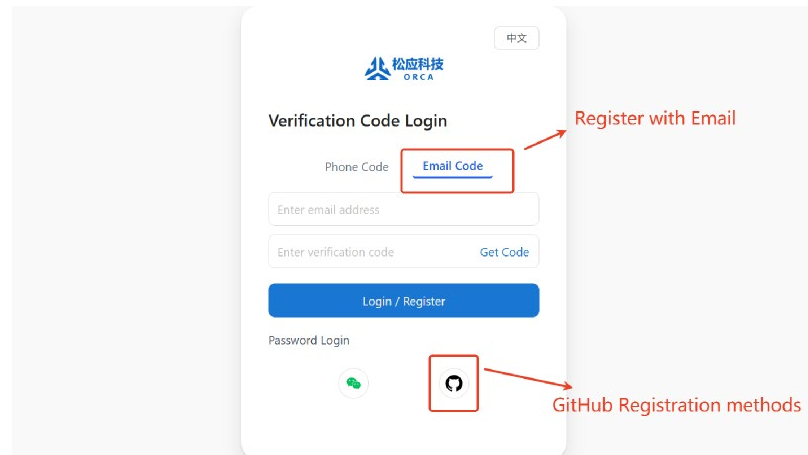
Standalone installable software with one-click rapid deployment.

Ubuntu installation guide

Windows installation guide (include pip/exe installation and how to set up the required environment)

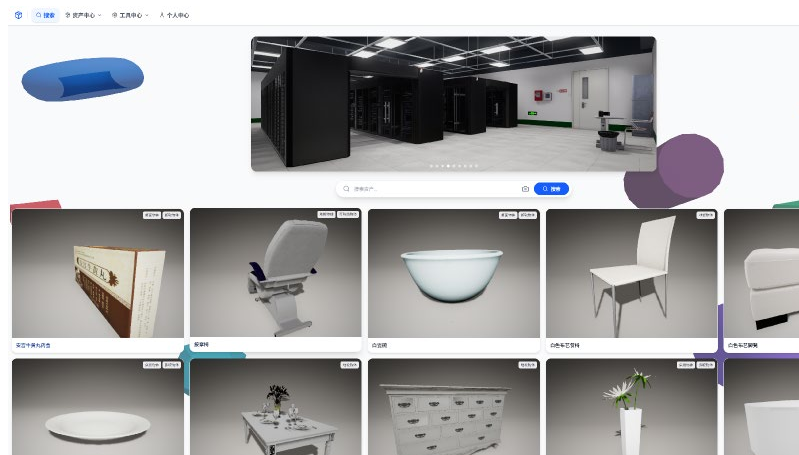
Step 2

Register Orca Asset Library Account



First thing

Click the link below and use your Email or GitHub account to sign up or log in. Signing up for an Asset Library account is a key step. The Orca Asset Library has thousands of 3D assets with physical properties to use with OrcaLab. <https://simassets.orca3d.cn/>



Second thing

After registration, you'll be logged in automatically, and the asset library interface will pop up. In the asset library, you can freely browse and subscribe to the asset content you're interested in.

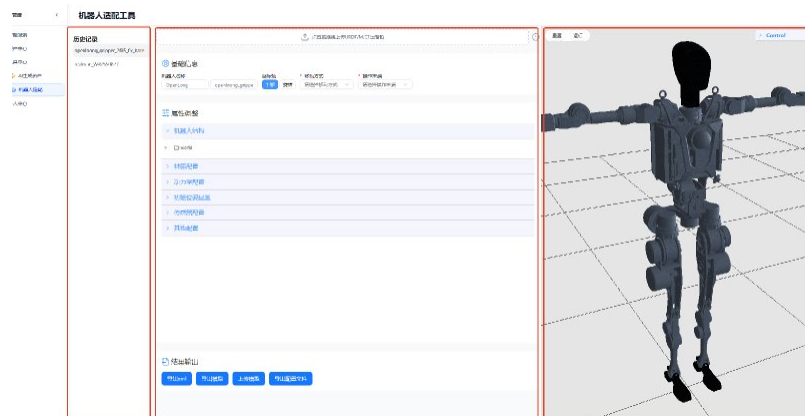
Step 2

Get Familiar with the Orca asset library



AI Asset Generation Tool

The Orca asset library has thousands of assets you can browse, subscribe to, and sync with the OrcaLab software. But if you can't find the asset you want here, why not try the AI asset generation feature, which lets you create the corresponding asset just from a text description.



Robot Adaptation Tool

At the same time, you can also upload your robot's URDF/MJCF model to this tool to use it with OrcaLab. Here, you can see the robot's basic information, adjust its properties, and perform operations like outputting results.

Step 3

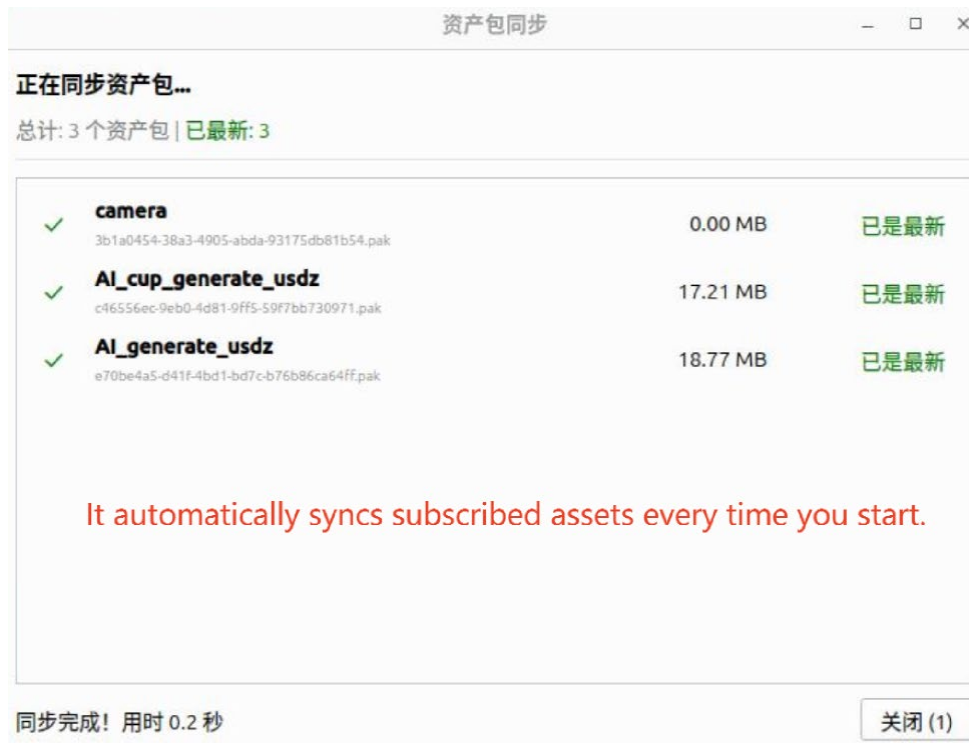
Run the OrcaLab

When finished installing OrcaLab and registering with the Orca asset library. Now you can run the OrcaLab, get familiar with the instructions, and master the basic operations.

You can choose to start OrcaLab using a shortcut or the command line tool. If you use the command line tool, you need to follow these steps:

```
# Enter the OrcaLab conda environment  conda  
activate orcalab  # Run OrcaLab  orcalab
```

Every time you start OrcaLab, it syncs the assets you've subscribed to in the asset library.

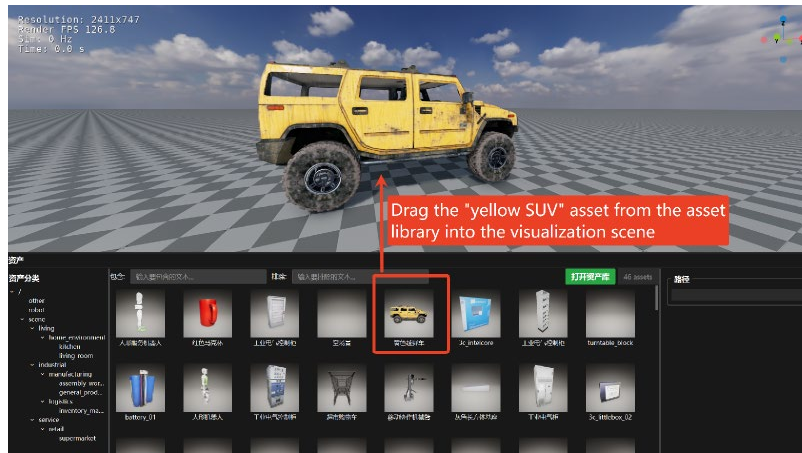


OrcaLab GUI Interface



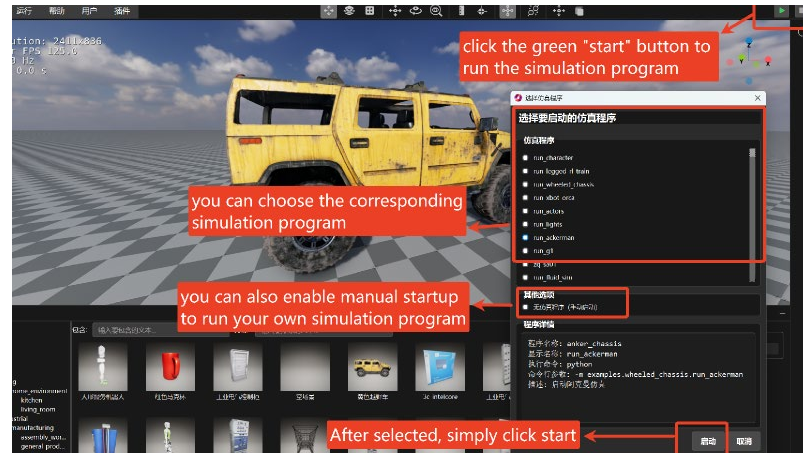
Launch the OrcaLab app, and the interface of the backend application looks like this after startup. OrcaLab supports flexible layouts. The default layout is divided into top, middle, and bottom layers, with a total of five modules: the menu bar, outline bar, view bar, editing bar, and asset bar. You can click on 'GUI Interface' to get more information about the UI.

Run a Simulation



Scene Setup

You can drag subscription assets from the asset library at the bottom of the interface into the visual scene above to set up your scene.



Quick Start Simulation Guide

After setting up the scene, you can start the simulation program to see how the objects change. The linked document has lots of simulation case tutorials you can learn from. These cases can help you get started faster and understand simulations.

Please continue exploring

The above is just a brief introduction to OrcaLab. You can continue reading our documentation, which has many simulation examples to help you get started with OrcaLab faster and understand more about remote operation, reinforcement learning, and data collection and synthesis.

<https://docs.orca3d.cn/en/#/README>

<https://github.com/openverse-orca>



ORCA Lab



ORCA

Songying Technology